Week 3

Chapter 3: Energy Bands, Electrons and Holes

Streetman and Banerjee, 'Solid State Electronic Devices,' 6th ed., Pearson, 2006.

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New Topics

- Equilibrium Carrier Concentrations
  - Density of States
  - Fermi function
  - Fermi level
  - Effective Mass
  - Non-degeneracy
  - Effective density of states
  - Intrinsic Carrier concentration
  - Intrinsic Fermi level
  - Intrinsic / Extrinsic

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A Brief Visit to the Quantum World

- The quantities that tie together our concepts of energy and electron or hole density are
  - Effective mass, m*
  - Density of states, $g_c(E)$ or $g_v(E)$.
- These quantities originate from a quantum description of the electronic states of a crystal.
- To understand them, we need to take a brief walk through quantum mechanics.

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Electrons as Waves

- The Schrödinger equation for a free particle is:

$$\frac{-\hbar^2}{2m}\nabla^2\Psi = E\Psi$$

- The solution is a plane wave:

$$\Psi_k(r) = \frac{e^{i\mathbf{k}\cdot\mathbf{r}}}{\sqrt{V}}$$

(k is wavevector)
(V is normalization volume)

- Putting this solution back into Schrödinger's Eq., gives:

$$\frac{\hbar^2 k^2}{2m}\Psi_k = E_k\Psi_k \Rightarrow E_k = \frac{\hbar^2 k^2}{2m}$$

- The 'states' $\Psi_k(r)$ are labeled by the wavevector k.

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Plane-Wave States

- The plane-wave states and energies are labeled by their wavevector k, Ψ_k and $E_k = \dfrac{\hbar^2 k^2}{2m^*}$
- In a crystal, the electrons and holes near the band edges act like free particles, except with an effective mass, m*.
- If we apply a force F, they accelerate as F/m* Instead of F/m.
- Usually m* is less than m.

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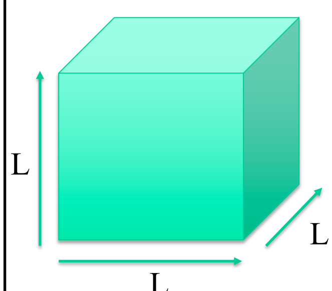
Normalization Volume



Sum over states → sum over k points.

Convert sum over k points to integral:

$$\frac{2}{V}\sum_{\boldsymbol{k}} = \frac{2}{V}\frac{V}{(2\pi)^3}\int dk_x dk_y dk_z$$

V=L^3

$$\Psi(\boldsymbol{r}) = \frac{e^{i\boldsymbol{k}\cdot\boldsymbol{r}}}{\sqrt{V}} = \frac{e^{ik_x x}e^{ik_y y}e^{ik_z z}}{\sqrt{V}}$$

Periodic Boundary Conditions

$$k_x = n_x \frac{2\pi}{L_x}$$

$$k_y = n_y \frac{2\pi}{L_y}$$

$$k_z = n_z \frac{2\pi}{L_z}$$

n_x, n_y, n_z are integers.

$L \longrightarrow \infty$

$\frac{2\pi^3}{V}$ is the volume in k-space associated with each k-point.

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Density of States

- To calculate physical properties like the electron density, n, we need to integrate over the electron states labeled by the 3D wavevector **k**.
- i.e. integrals like this:

spin

$$\frac{2}{(2\pi)^3}\int dk_x dk_y dk_z f(k) = \frac{2}{(2\pi)^3}\int_0^{2\pi} d\phi \int_0^{\pi} d\phi \sin(\phi) \int_0^{\infty} dk k^2 f(k) = \frac{2}{(2\pi)^3}(4\pi)\int_0^{\infty} dk k^2 f(k)$$

Inverse Fourier Transform

- Now I change variables from k to E using $E = \dfrac{\hbar^2 k^2}{2m^*}$:

$$dE = \frac{\hbar^2 k}{m^*} dk \quad \text{and} \quad k = \sqrt{2m^* E}/\hbar \ \Rightarrow dk\ k^2 = dE\ \frac{m^* \sqrt{2m^* E}}{\hbar^3}$$

$$\frac{2}{(2\pi)^3}\int dk_x dk_y dk_z f(k) = \int_0^{\infty} dE \boxed{\underbrace{\frac{m^* \sqrt{2m^* E}}{\pi^2 \hbar^3}}_{g(E)}} f(E)$$

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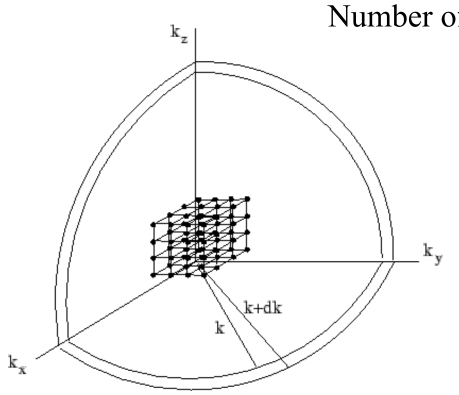
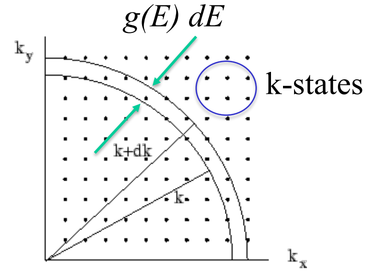
DOS – State Counting

Number of states between E and $E + dE$.





$$\frac{2}{(2\pi)^3}\int dk_x dk_y dk_z f(k) = \int_0^{\infty} \boxed{dE \underbrace{\frac{m^* \sqrt{2m^* E}}{\pi^2 \hbar^3}}_{g(E)}} f(E)$$

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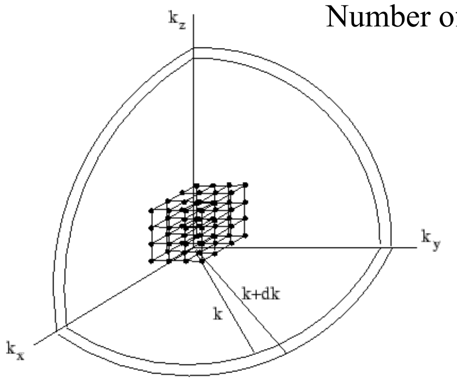
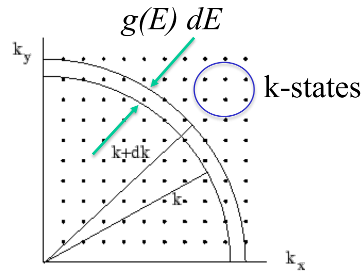
Number of states between E and $E + dE$.





$$g(E) = \frac{dn}{dE} = \frac{dn}{dk}\frac{dk}{dE}$$

$$n(k) = \frac{2}{(2\pi)^3}\frac{4}{3}\pi k^3 \qquad \frac{dn}{dk} = k^2/\pi^2$$

$$g(E) = \frac{dn}{dk}\frac{dk}{dE} = \frac{m^* k}{\pi^2\hbar^2}$$

$$E = \frac{\hbar^2 k^2}{2m*} \ ; \quad k = \frac{\sqrt{2m^* E}}{\hbar}$$

$$\boxed{g(E) = \frac{m^*\sqrt{2m^* E}}{\pi^2\hbar^3}}$$

$$\frac{dk}{dE} = \left(\frac{dE}{dk}\right)^{-1} = \left(\frac{\hbar^2 k}{m^*}\right)^{-1} = \frac{m^*}{\hbar^2 k}$$

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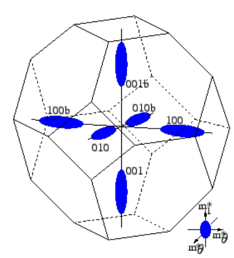
Density of States



Iso-energy surfaces of Si near the conduction band edge.

- The conduction band of Si in k-space, is 6-fold degenerate. It sits at 6 equivalent points in k-space.
- So we need to multiply by this degeneracy, M_c:

$$g_c(E) = M_c \frac{m^*\sqrt{2m^* E}}{\pi^2\hbar^3}$$

- Furthermore, for the elipse along the x-axis, $E(k) = \frac{\hbar^2 k_x^2}{2m_l} + \frac{\hbar^2\left(k_y^2 + k_z^2\right)}{2m_t}$ i.e. there are 2 masses, longitudinal and transverse.
- To make the ellipses look like spheres, we define

$$m^* = (m_l m_t m_t)^{1/3}$$

- Finally Streetman and Pierret get rid of M_c by redefining m* as

$$m_n^* = \left(M_c\right)^{2/3}(m_l m_t m_t)^{1/3} = 6^{2/3}(0.98\times 0.19\times 0.19)^{1/3} m_0 = 1.08 m_0$$

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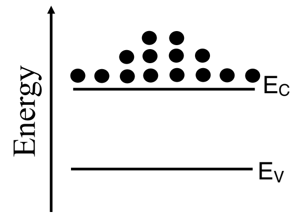
Density of States Effective Mass

- Density of states m*:.
  - Conduction band DOS m*: m_n^*
  - Valence band DOS m*: m_p^*



Material	DOS m_n^*/m_0	DOS m_p^*/m_0
Si	1.08	0.81
Ge	0.55	0.36
GaAs	0.066	0.52

m_0 = free electron mass.

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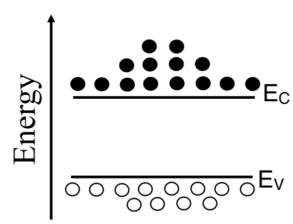
Carrier Concentration

- Density of states (DOS):
  - Density of available states in the conduction band ($g_c(E)$) and valence band ($g_v(E)$).
  - Function of energy $\sqrt{E}$.



$$g_c(E) = \frac{m_n^* \sqrt{2m_n^*(E - E_C)}}{\pi^2 \hbar^3} \quad (E \geq E_C)$$

Conduction band DOS

$$g_v(E) = \frac{m_p^* \sqrt{2m_p^*(E_V - E)}}{\pi^2 \hbar^3} \quad (E_V \geq E)$$

Valence band DOS

m* = density of states effective mass

$\hbar$ = Reduced Planck's constant

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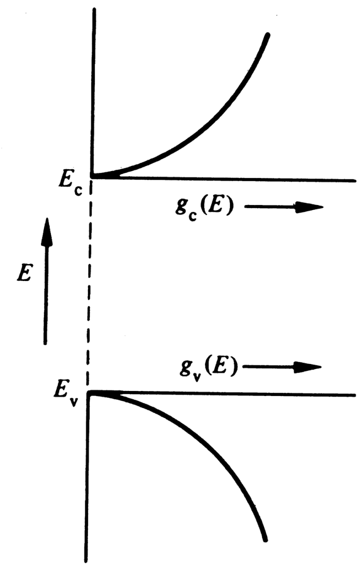
Density of States

- Square root of energy.

$$g_c(E) = \frac{m_n^* \sqrt{2m_n^*(E - E_C)}}{\pi^2 \hbar^3} \quad (E \geq E_C)$$

$$g_v(E) = \frac{m_p^* \sqrt{2m_p^*(E_V - E)}}{\pi^2 \hbar^3} \quad (E_V \geq E)$$



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Fermi Function

- In equilibrium, the probability that a state is occupied with an electron is given by the Fermi function *f(E)*:

$$f(E) = \frac{1}{1 + \exp\left(\dfrac{E - E_F}{k_B T}\right)}$$

- E_F = Fermi Energy.
- k_B = Boltzmann's constant, 8.617e-5 (eV/K).
- T = temperature (K).
- E = Energy of the electron.
- $f(E << E_F) = 1$.
- $f(E = E_F) = ½$.
- $f(E >> E_F) = 0$.

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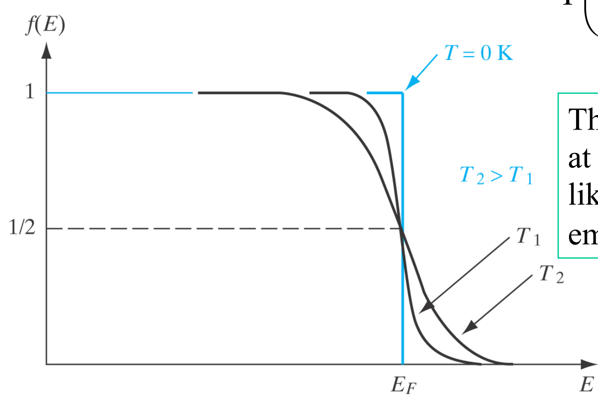
Fermi Function

- The probability that a state is occupied with an electron

$$f(E) = \frac{1}{1+\exp\left(\frac{E-E_F}{k_B T}\right)}$$



The Fermi level is the energy at which a state is equally likely to be occupied or empty. $f(E_F) = ½$.

B. G. Streetman and S. K. Banerjee, Solid State Electronic Devices, 6th ed. Pearson, 2006.

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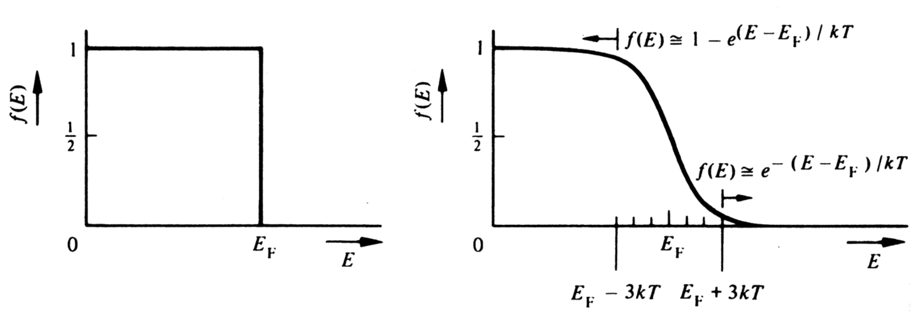
Fermi Function

$$\boxed{f(E) = \frac{1}{1+\exp\left(\frac{E-E_F}{k_B T}\right)}}$$



(a) $T \to 0$ K (b) $T > 0$ K

R. F. Pierret, Semiconductor Device Fundamentals, Addison-Wesley, 1996.

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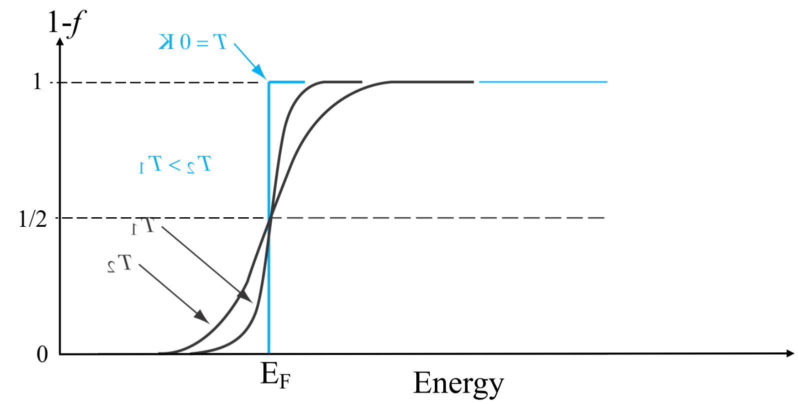
Fermi Function

- The probability that a state is occupied with a hole is [1 – $f(E)$].



B. G. Streetman and S. K. Banerjee, Solid State Electronic Devices, 6th ed. Pearson, 2006.

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Electron and Hole Density

- Multiply the density of states times the probability that the state is occupied and integrate.

$$n = \int_{E_C}^{\infty} g_c(E) f(E) dE$$

$$p = \int_{-\infty}^{E_V} g_v(E)\left(1 - f(E)\right) dE$$

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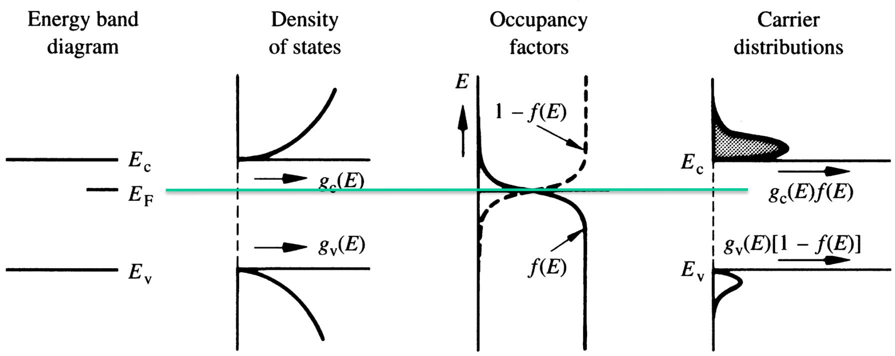
Electron and Hole Density

- Multiply the density of states times the probability that the state is occupied and integrate.



(a) E_F above midgap

R. F. Pierret, Semiconductor Device Fundamentals, Addison-Wesley, 1996.

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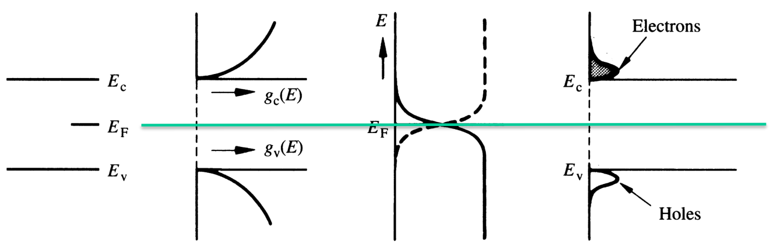
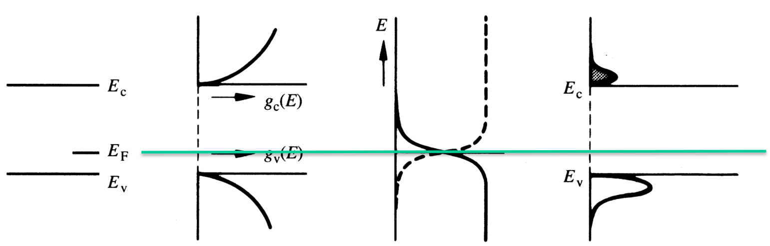


(b) E_F near midgap



(c) E_F below midgap

R. F. Pierret, Semiconductor Device Fundamentals, Addison-Wesley, 1996.

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Electron Density

$$n = \int_{E_C}^{\infty} g_c(E) f(E) dE$$

$$n = \frac{m_n^* \sqrt{2m_n^*}}{\pi^2 \hbar^3} \int_{E_C}^{\infty} \frac{\sqrt{E - E_C}}{1 + e^{(E-E_F)/k_B T}} dE = N_C \frac{2}{\sqrt{\pi}} F_{1/2}(\eta_c)$$

$F_{1/2}$ = Fermi-Dirac integral of order ½.

$\eta_c = (E_F - E_C) / k_B T$

This can be computed numerically or looked up in tables or approximated.

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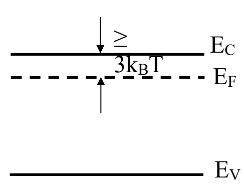
Electron Density

$$n = \frac{m_n^* \sqrt{2m_n^*}}{\pi^2 \hbar^3} \int_{E_C}^{\infty} \frac{\sqrt{E - E_C}}{1 + e^{(E-E_F)/k_B T}} dE = N_C \frac{2}{\sqrt{\pi}} F_{1/2}(\eta_c)$$

Non-degenerate



For $E_C - E_F \geq 3\ k_B T$,

$$f(E) = \frac{1}{1 + e^{(E-E_F)/k_B T}} \approx e^{-(E-E_F)/k_B T}$$

since $e^3 = 20 >> 1$

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Electron Density

$$n = \frac{m_n^* \sqrt{2m_n^*}}{\pi^2 \hbar^3} \int_{E_C}^{\infty} \frac{\sqrt{E - E_C}}{1 + e^{(E-E_F)/k_B T}}\, dE = N_C \frac{2}{\sqrt{\pi}} F_{1/2}(\eta_c)$$

Non-degenerate

$$f(E) \approx e^{-(E-E_F)/k_B T}$$

$$n \approx \frac{m_n^* \sqrt{2m_n^*}}{\pi^2 \hbar^3} \int_{E_C}^{\infty} \sqrt{E - E_C}\; e^{-(E-E_F)/k_B T}\, dE$$

$$\boxed{n \approx N_C\, e^{-(E_C - E_F)/k_B T}}$$

where

$$\boxed{N_C = 2\left[\frac{m_n^* k_B T}{2\pi\hbar^2}\right]^{\frac{3}{2}}}$$

N_C = Effective density of conduction band states.

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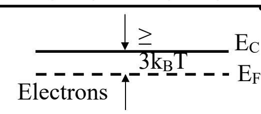
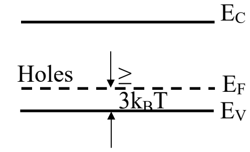
Non-Degenerate Electron and Hole Density

Non-degenerate:

- Electrons: $E_C - E_F \geq 3\ k_B T$
- Holes: $E_F - E_V \geq 3\ k_B T$





$$\boxed{n \approx N_C\, e^{-(E_C - E_F)/k_B T}} \qquad \boxed{N_C = 2\left[\frac{m_n^* k_B T}{2\pi\hbar^2}\right]^{\frac{3}{2}}}$$

$$\boxed{p \approx N_V\, e^{-(E_F - E_V)/k_B T}} \qquad \boxed{N_V = 2\left[\frac{m_p^* k_B T}{2\pi\hbar^2}\right]^{\frac{3}{2}}}$$

N_C = Effective density of conduction band states.

N_V = Effective density of valence band states.

$$\boxed{N_{C,V} = \left(2.510 \times 10^{19} / \mathrm{cm}^3\right)\left(m_{n,p}^* / m_0\right)^{3/2}} \quad (\mathrm{T} = 300\mathrm{K})$$

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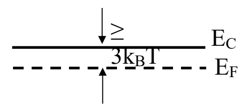
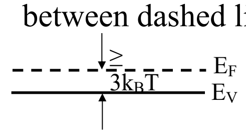
Non-Degenerate

- Know this.



Non-degenerate iff E_F lies in this region between dashed lines.



Otherwise, it is <u>degenerate</u>.

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Electron and hole density, n and p

- Know this.

$$n = N_C \, e^{-(E_C - E_F)/k_B T}$$

$$p = N_V \, e^{-(E_F - E_V)/k_B T}$$

$$N_C = 2\left[\frac{m_n^* k_B T}{2\pi\hbar^2}\right]^{3/2}$$

$$N_V = 2\left[\frac{m_p^* k_B T}{2\pi\hbar^2}\right]^{3/2}$$

$$N_{C,V} = \left(2.510 \times 10^{19} / \mathrm{cm}^3\right)\left(m_{n,p}^* / m_0\right)^{3/2}$$ (T = 300K)

$$N_{C,V} = \left(2.510 \times 10^{19} / \mathrm{cm}^3\right)\left(m_{n,p}^* / m_0\right)^{3/2} \left(T/300\right)^{3/2}$$

R. F. Pierret, Semiconductor Device Fundamentals, Addison-Wesley, 1996.

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n and p in Terms of the Intrinsic Carrier Concentration, n_i and Intrinsic Fermi Level, E_i

$$\boxed{n_i = N_C\, e^{-(E_C - E_i)/k_B T}}$$

$$\boxed{p_i = N_V\, e^{-(E_i - E_V)/k_B T} = n_i}$$

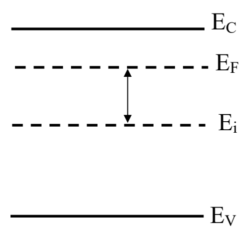
$$N_C = n_i e^{(E_C - E_i)/k_B T}$$

$$\boxed{n = N_C\, e^{-(E_C - E_F)/k_B T}}$$

$$\boxed{p = N_V\, e^{-(E_F - E_V)/k_B T}}$$

$$\boxed{n = n_i\, e^{(E_F - E_i)/k_B T}}$$

$$\boxed{p = n_i\, e^{(E_i - E_F)/k_B T}}$$



n increases exponentially as (E_F–E_i) increases.

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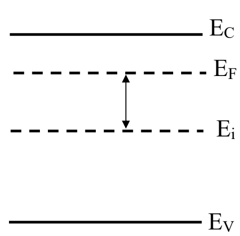
n and p

- Know this

$$\boxed{n = n_i\, e^{(E_F - E_i)/k_B T}}$$

$$\boxed{p = n_i\, e^{(E_i - E_F)/k_B T}}$$



n increases exponentially as (E_F–E_i) increases.

p increases exponentially as (E_i–E_F) increases.

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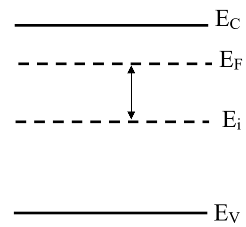
Two Equations for n and p

- Know this

$$\boxed{\begin{aligned} n &= n_i\, e^{(E_F-E_i)/k_BT} \\ p &= n_i\, e^{(E_i-E_F)/k_BT} \end{aligned}}$$



$$\boxed{\begin{aligned} n &= N_C\; e^{-(E_C-E_F)/k_BT} \\ p &= N_V\; e^{-(E_F-E_V)/k_BT} \end{aligned}}$$

Valid for non-degenerate semiconductor.

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Aside - How to Avoid MKS Units

$$N_C = 2\left[\frac{m_n^* k_B T}{2\pi\hbar^2}\right]^{3/2}$$

$$m_0c^2 = 0.511\times10^6\ \text{eV} = 0.511\ \text{MeV}$$

$$c = 3\times10^{18}\ \overset{\circ}{\text{A}}/\text{s (angstroms / s)}$$

$$\hbar c = 1973\ \text{eV}\ \overset{\circ}{\text{A}}$$

$$1\ \overset{\circ}{\text{A}} = 10^{-8}\ \text{cm}$$

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How to Avoid MKS Units

$$N_C = 2\left[\frac{m_n^* k_B T}{2\pi\hbar^2}\right]^{3/2} = 2\left[\frac{m_n^* c^2 k_B T}{2\pi\hbar^2 c^2}\right]^{3/2}$$

$m_0 c^2 = 0.511\times10^6 \text{ eV} = 0.511 \text{ MeV}$

$c = 3\times10^{18}\ \text{Å/s (angstroms / s)}$

$\hbar c = 1973 \text{ eV Å}$

$1 \text{ Å} = 10^{-8} \text{ cm}$

$$= 2\left[\frac{1.08 * 0.511\text{e}6 \text{ (eV)} * 0.02585 \text{ (eV)}}{2\pi * (1973)^2 \left(\text{eV} \cdot \text{Å}\right)^2 \left(\dfrac{1 \text{ cm}}{10^8 \text{ Å}}\right)^2}\right]^{3/2} = 2.82\times10^{19} \text{ / cm}^3$$

T=300K

$$\boxed{N_{C,V} = \left(2.510\times10^{19} \text{ / cm}^3\right)\left(m_{n,p}^* / m_0\right)^{3/2} (T/300)^{3/2}}$$

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What is the Value of the Intrinsic Carrier Concentration, n_i?

$$n_i p_i = N_C\, e^{-(E_C - E_i)/k_B T} N_V\, e^{-(E_i - E_V)/k_B T}$$

$$n_i p_i = N_C N_V\, e^{-E_G/(k_B T)}$$

$$\boxed{n_i = p_i = \sqrt{N_C N_V}\; e^{-E_G/(2 k_B T)}}$$

Note the exponential dependence on bandgap and temperature.

Semi.	E_G (eV)	n_i (cm^{-3}) (T=300K)
Si	1.12	1.0e10
Ge	0.66	2.5e13
GaAs	1.42	2e6

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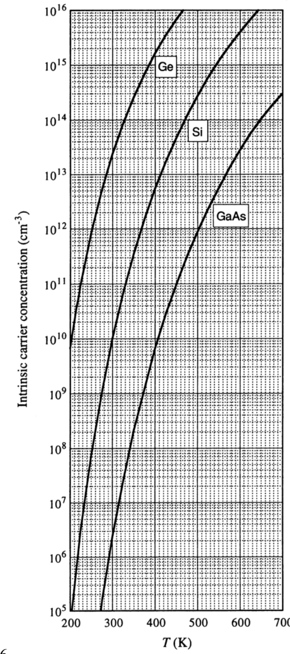
n_i(T)

- Exponential dependence on 1/T and the bandgap, E_G.

$$n_i = p_i = \sqrt{N_C N_V}\ \ e^{-E_G/(2k_BT)}$$



Si	
T(°C)	n_i(cm^{-3})
0	8.86×10^8
5	1.44×10^9
10	2.30×10^9
15	3.62×10^9
20	5.62×10^9
25	8.60×10^9
30	1.30×10^{10}
35	1.93×10^{10}
40	2.85×10^{10}
45	4.15×10^{10}
50	5.97×10^{10}
300 K	1.00×10^{10}

GaAs	
T(°C)	n_i(cm^{-3})
0	1.02×10^5
5	1.89×10^5
10	3.45×10^5
15	6.15×10^5
20	1.08×10^6
25	1.85×10^6
30	3.13×10^6
35	5.20×10^6
40	8.51×10^6
45	1.37×10^7
50	2.18×10^7
300 K	2.25×10^6

R. F. Pierret, Semiconductor Fundamentals, Addison Wesley Longman, 1996.

Figure 2.20

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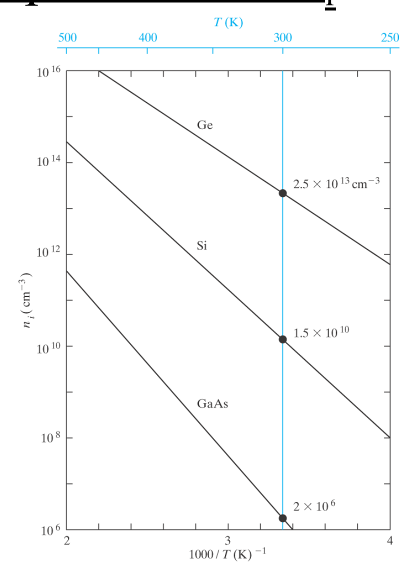
Temperature Dependence of n_i

Exponential term dominates.



$$n_i = \sqrt{N_C N_V}\ \ e^{-E_G/(2k_BT)}$$

$$n_i = 2(m_n^* m_p^*)^{3/4}\left(\frac{2\pi k_B}{\hbar^2}\right)^{3/2} T^{3/2} e^{-E_G/(2k_BT)}$$

B. G. Streetman and S. K. Banerjee, Solid State Electronic Devices, 6th ed. Pearson, 2006.

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What is the Value of E_i?

$$n_i = N_C\, e^{-(E_C - E_i)/k_BT} = N_V\, e^{-(E_i - E_V)/k_BT}$$

$$\frac{N_C}{N_V} = e^{-2E_i/(k_BT)} e^{+(E_C+E_V)/k_BT}$$

$$\ln\left(\frac{N_C}{N_V}\right) = [E_C + E_V - 2E_i]/k_BT$$

$$E_i = \frac{E_C + E_V}{2} + \frac{k_BT}{2}\ln\left(\frac{N_V}{N_C}\right)$$

$$E_i = \frac{E_C + E_V}{2} + \frac{3}{4}k_BT\ln\left(\frac{m_p^*}{m_n^*}\right) = \frac{E_C + E_V}{2} - 0.007\ \text{(eV)}$$

Almost at midgap.

Si @ T=300K

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np Product

Multiply these together:

$$\boxed{n = n_i\, e^{(E_F - E_i)/k_BT}}$$

$$\boxed{p = n_i\, e^{(E_i - E_F)/k_BT}}$$

$$np = n_i^2\, e^{(E_F - E_i)/k_BT} e^{(E_i - E_F)/k_BT}$$

$$\boxed{np = n_i^2}$$

In equilibrium,

$$\boxed{np = n_i^2}$$

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np product

- Know this
- In equilibrium:

$$\boxed{np = n_i^2}$$

R. F. Pierret, Semiconductor Fundamentals, Addison Wesley Longman, 1996.

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Charge Neutrality

- A uniformly doped semiconductor is everywhere charge neutral.

$$q\left(p - n + N_D^+ - N_A^-\right) = 0$$ Net charge = 0.

$$p - n + N_D^+ - N_A^- = 0$$

$$p - n + N_D - N_A = 0$$ Assume complete ionization of dopants

$$\frac{n_i^2}{n} - n + N_D - N_A = 0$$ $np = n_i^2$

$$n^2 - n(N_D - N_A) - n_i^2 = 0$$ Quadratic equation for n or p.

$$\boxed{\begin{aligned} n &= \frac{N_D - N_A}{2} + \sqrt{\left(\frac{N_D - N_A}{2}\right)^2 + n_i^2} \\ p &= \frac{N_A - N_D}{2} + \sqrt{\left(\frac{N_A - N_D}{2}\right)^2 + n_i^2} \end{aligned}}$$

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Carrier Concentrations

$$n = \frac{N_D - N_A}{2} + \sqrt{\left(\frac{N_D - N_A}{2}\right)^2 + n_i^2}$$

$$p = \frac{N_A - N_D}{2} + \sqrt{\left(\frac{N_A - N_D}{2}\right)^2 + n_i^2}$$

- $N_D >> N_A, n_i$
  - $n \approx N_D$
  - $p \approx n_i^2 / N_D$
- $N_A >> N_D, n_i$
  - $p \approx N_A$
  - $n \approx n_i^2 / N_A$

- Compensation
  - $N_A \approx N_D$
  - $n \approx p \approx n_i$
  - The p-type doping compensates the n-type doping.

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Carrier Concentrations

$$\boxed{\begin{aligned} n &\simeq N_D \\ p &\simeq n_i^2/N_D \end{aligned}}$$

$N_D \gg N_A, \quad N_D \gg n_i$
(nondegenerate, total ionization)

$$\boxed{\begin{aligned} p &\simeq N_A \\ n &\simeq n_i^2/N_A \end{aligned}}$$

$N_A \gg N_D, \quad N_A \gg n_i$
(nondegenerate, total ionization)

Practical Carrier Conc. Relationships

R. F. Pierret, Semiconductor Fundamentals, Addison Wesley Longman, 1996.

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Where is E_F?

- Normal case of $N_D >> n_i, N_A$, $n = N_D$.

$$N_D = n_i\, e^{(E_F - E_i)/k_B T}$$

$$\boxed{E_F - E_i = k_B T \ln(N_D / n_i)}$$

$$n = n_i\, e^{(E_F - E_i)/k_B T}$$

- $N_A >> n_i, N_D$, $p = N_A$.

$$N_A = n_i\, e^{(E_i - E_F)/k_B T}$$

$$\boxed{E_i - E_F = k_B T \ln(N_A / n_i)}$$

$$p = n_i\, e^{(E_i - E_F)/k_B T}$$

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E_F versus doping

$$E_F - E_i = k_B T \ln(N_D / n_i)$$

$$E_i - E_F = k_B T \ln(N_A / n_i)$$

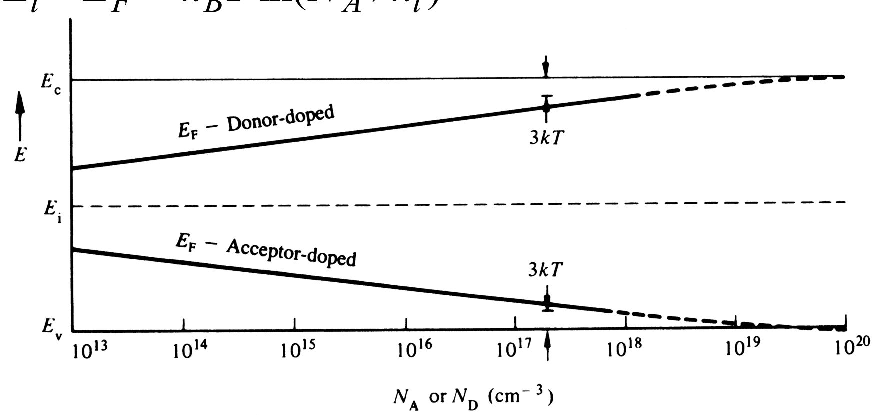

Fig. 2.21 (E_F vs. Dopant Conc.)

R. F. Pierret, Semiconductor Fundamentals, Addison Wesley Longman, 1996.

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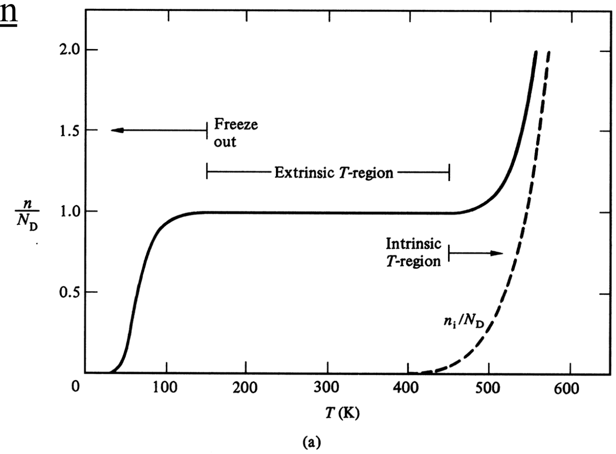
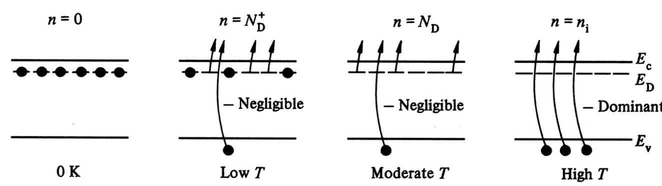
T-Dependence of n





R. F. Pierret, Semiconductor Fundamentals, Addison Wesley Longman, 1996.

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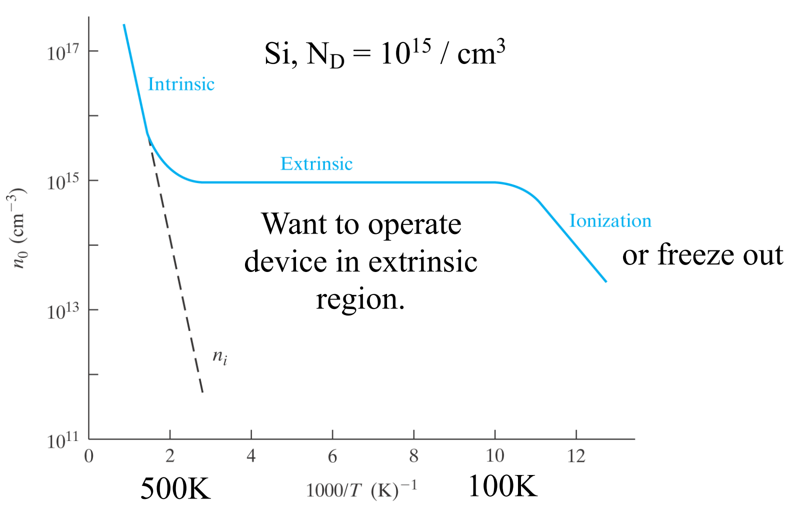
Temperature Dependence of the electron density, n



B. G. Streetman and S. K. Banerjee, Solid State Electronic Devices, 6th ed. Pearson, 2006.

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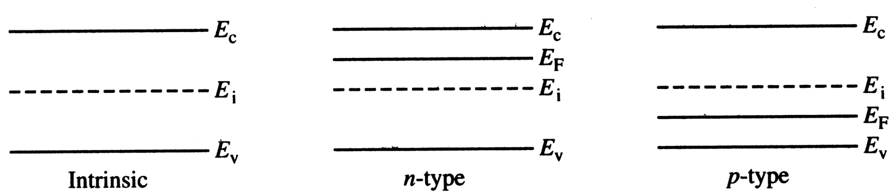
Band diagram

- Draw the band diagram for an n-type semiconductor

- Draw the band diagram for an p-type semiconductor

E_x

45



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Retrieval Session

Which of the following conditions would give rise to such a band diagram? Explain.

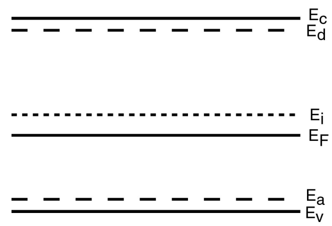
(i) Very high temperature ($kT = E_G$)

(ii) High acceptor doping

(iii) Low acceptor doping

(iv) High donor doping

(v) Low donor doping



since E_F is slightly below E_i.

What would (i), (ii), (iv) or (v) imply?

Retrieval Session

Which of the following conditions would give rise to such a band diagram? Explain.

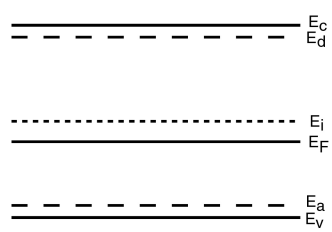
(i) Very high temperature ($kT = E_G$)

(ii) High acceptor doping

(iii) Low acceptor doping

(iv) High donor doping

(v) Low donor doping



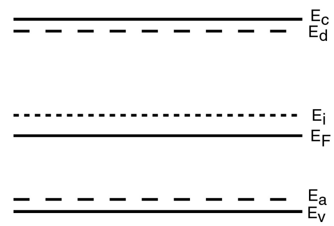
since E_F is slightly below E_i.

(i) ➔ intrinsic, $E_F = E_i$, (ii) ➔ E_F near E_V, (iv) ➔ E_F near E_C, (v) ➔ E_F slightly above E_i.

What do the lines E_d and E_a tell you?

Retrieval Session

Which of the following conditions would give rise to such a band diagram? Explain.



(i) Very high temperature ($kT = E_G$)

(ii) High acceptor doping

(iii) Low acceptor doping

since E_F is slightly below E_i.

(iv) High donor doping

(v) Low donor doping

(i) ➔ intrinsic, $E_F = E_i$, (ii) ➔ E_F near E_V, (iv) ➔ E_F near E_C, (v) ➔ E_F slightly above E_i.

What do the lines E_d and E_a tell you?

Nothing relevant to this problem.

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Retrieval Session

- In Si (T=300K), approximately what is n_i?

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Retrieval Session

- In Si (T=300K), approximately what is n_i?

  $n_i = 1.45 \times 10^{10}$ / cm^3 $\approx 10^{10}$ / cm^3 .

- In Si (T=300K), $N_D = 10^{15}$ / cm^3.
  - What is n?
  - What is p?

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Retrieval Session

- In Si (T=300K), approximately what is n_i?

  $n_i = 1.45 \times 10^{10}$ / cm^3 $\approx 10^{10}$ / cm^3 .

- In Si (T=300K), $N_D = 10^{15}$ / cm^3.
  - What is n? $n = N_D = 10^{15}$ / cm^3.
  - What is p? $p = \frac{n_i^2}{N_D} = \frac{2.1 \times 10^{20}}{10^{15}} = 2.1 \times 10^5 \ / \mathrm{cm}^3$

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Retrieval Session

- Given that $N_C = N_V$, $E_V = 0$, and $E_C = 1$ eV, what is E_i?

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Retrieval Session

- Given that $N_C = N_V$, $E_V = 0$, and $E_C = 1$ eV, what is E_i?

$$E_i = \frac{E_C + E_V}{2} + \frac{k_B T}{2} \cancelto{0}{\ln\left(\frac{N_V}{N_C}\right)}$$

$$E_i = \frac{E_C + E_V}{2} + \frac{3}{4} k_B T \ln\left(\frac{m_p^*}{m_n^*}\right)$$

$\boxed{E_i = 0.5 \text{ eV.}}$

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Retrieval Session

- T=300K. Given that $E_C - E_F = 0.2$ eV and $N_D = 10^{15}$ / cm^3, what is N_C?

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Retrieval Session

- T=300K. Given that $E_C - E_F = 0.2$ eV and $N_D = 10^{15}$ / cm^3, what is N_C?

$$n = N_D$$

$$n = N_D = N_C \ e^{-(E_C - E_F)/k_B T}$$

$$N_C = N_D e^{(E_C - E_F)/k_B T} = 10^{15} e^{0.2/0.026} = 2.19 \times 10^{18} \ / \ \text{cm}^3$$

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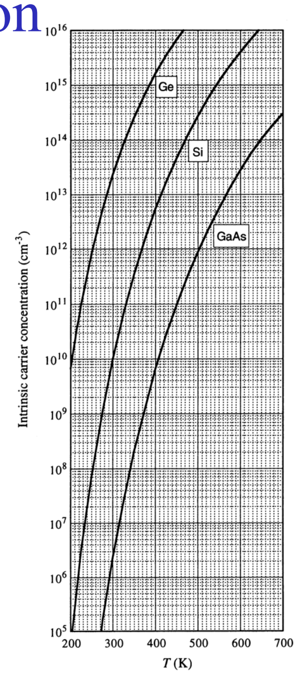
Retrieval Session

- Si is doped $N_D = 10^{14}/cm^3$.
  - When do I need to start worrying about the formula $n = N_D$?



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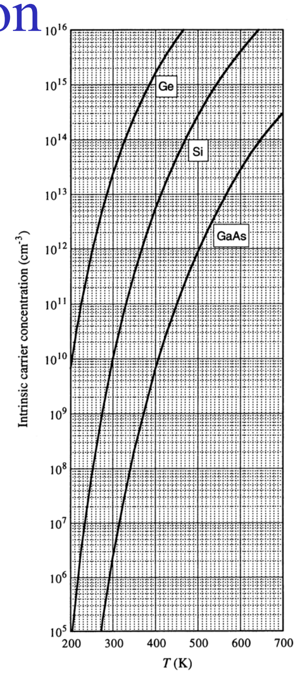
Retrieval Session

- Si is doped $N_D = 10^{14}/cm^3$.
  - As T rises above 300K, when do I need to start worrying about the formula $n = N_D$?

When n_i becomes comparable to N_D.

$$n = \frac{N_D - N_A}{2} + \sqrt{\left(\frac{N_D - N_A}{2}\right)^2 + n_i^2}$$

$$p = \frac{N_A - N_D}{2} + \sqrt{\left(\frac{N_A - N_D}{2}\right)^2 + n_i^2}$$



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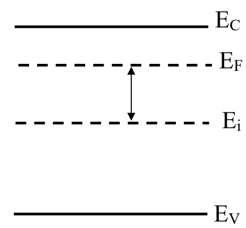
Two Equations for n and p

- Know this

$$n = n_i \, e^{(E_F - E_i)/k_B T}$$
$$p = n_i \, e^{(E_i - E_F)/k_B T}$$



$$n = N_C \; e^{-(E_C - E_F)/k_B T}$$
$$p = N_V \; e^{-(E_F - E_V)/k_B T}$$

Valid for non-degenerate semiconductor.

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Review

- Intrinsic Carrier Concentration, n_i

$$n_i = p_i = \sqrt{N_C N_V} \;\; e^{-E_G/(2k_B T)}$$

- Intrinsic Fermi level, E_i

$$E_i = \frac{E_C + E_V}{2} + \frac{k_B T}{2} \ln\left(\frac{N_V}{N_C}\right)$$
$$E_i = \frac{E_C + E_V}{2} + \frac{3}{4} k_B T \ln\left(\frac{m_p^*}{m_n^*}\right)$$

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Review

• np Product

$$n = n_i \, e^{(E_F - E_i)/k_BT}$$

$$p = n_i \, e^{(E_i - E_F)/k_BT}$$

$$np = n_i^2 \; e^{(E_F - E_i)/k_BT} e^{(E_i - E_F)/k_BT}$$

In equilibrium,

$$\boxed{np = n_i^2}$$

Review

• Carrier concentration

In general:

$$n = \frac{N_D - N_A}{2} + \sqrt{\left(\frac{N_D - N_A}{2}\right)^2 + n_i^2}$$

$$p = \frac{N_A - N_D}{2} + \sqrt{\left(\frac{N_A - N_D}{2}\right)^2 + n_i^2}$$

Usually:

$$\boxed{\begin{aligned} n &\simeq N_D \\ p &\simeq n_i^2/N_D \end{aligned}}$$

$N_D \gg N_A, \quad N_D \gg n_i$
(nondegenerate, total ionization)

$$\boxed{\begin{aligned} p &\simeq N_A \\ n &\simeq n_i^2/N_A \end{aligned}}$$

$N_A \gg N_D, \quad N_A \gg n_i$
(nondegenerate, total ionization)

Practical Carrier Concentration Relationships

R. F. Pierret, Semiconductor Fundamentals, Addison Wesley Longman, 1996.

Review

Summary of equations.

1. Density of states.
2. Fermi factor or Fermi-Dirac factor.
3. Fermi level.
4. Effective density of states (N_C, N_V).
5. Intrinsic carrier concentration, n_i.
6. Intrinsic Fermi level, E_i.
7. $np = n_i^2$.
8. $N_D >> n_i, N_A$: $n=N_D$, $p= n_i^2/ N_D$.
9. $N_A >> n_i, N_D$: $p=N_A$, $n=n_i^2/ N_A$.

Table 2.4 Carrier Modeling Equation Summary.

Density of States and Fermi Function

$$g_c(E) = \frac{m_n^*\sqrt{2m_n^*(E-E_c)}}{\pi^2\hbar^3}, \quad E \ge E_c$$

$$g_v(E) = \frac{m_p^*\sqrt{2m_p^*(E_v-E)}}{\pi^2\hbar^3}, \quad E \le E_v$$

$$f(E) = \frac{1}{1+e^{(E-E_F)/kT}}$$

Carrier Concentration Relationships

$$n = N_C \frac{2}{\sqrt{\pi}} F_{1/2}(\eta_c) \qquad N_C = 2\left[\frac{m_n^* kT}{2\pi\hbar^2}\right]^{3/2}$$

$$p = N_V \frac{2}{\sqrt{\pi}} F_{1/2}(\eta_v) \qquad N_V = 2\left[\frac{m_p^* kT}{2\pi\hbar^2}\right]^{3/2}$$

$$n = N_C e^{(E_F-E_c)/kT}$$

$$p = N_V e^{(E_v-E_F)/kT}$$

$$n = n_i e^{(E_F-E_i)/kT}$$

$$p = n_i e^{(E_i-E_F)/kT}$$

n_i, np-Product, and Charge Neutrality

$$n_i = \sqrt{N_C N_V}\, e^{-E_G/2kT} \qquad np = n_i^2 \qquad p - n + N_D - N_A = 0$$

n, p, and Fermi Level Computational Relationships

$$n = \frac{N_D-N_A}{2} + \left[\left(\frac{N_D-N_A}{2}\right)^2 + n_i^2\right]^{1/2} \qquad E_i = \frac{E_c+E_v}{2} + \frac{3}{4}kT\ln\left(\frac{m_p^*}{m_n^*}\right)$$

$$n \simeq N_D, \quad p \simeq n_i^2/N_D \qquad N_D \gg N_A, N_D \gg n_i$$

$$E_F - E_i = kT\ln(n/n_i) = -kT\ln(p/n_i)$$

$$p \simeq N_A, \quad n \simeq n_i^2/N_A \qquad N_A \gg N_D, N_A \gg n_i$$

$$E_F - E_i = kT\ln(N_D/n_i) \quad N_D \gg N_A, N_D \gg n_i$$

$$E_i - E_F = kT\ln(N_A/n_i) \quad N_A \gg N_D, N_A \gg n_i$$

R. F. Pierret, Semiconductor Fundamentals, Addison Wesley Longman, 1996.

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